

PAPER CODE: BCE-O633
SEMESTER EXAMINATION (MAY-2025)

CLASS: B.TECH.

SEMESTER: VI

SUBJECT: COMPUTER SCIENCE AND ENGINEERING

PAPER TITLE: INTRODUCTION TO DATA SCIENCE AND DESIGN THINKING

Time: 3 Hours

Max. Marks: 70

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any *five* questions in about 150 words each. Each question carries six marks. **(5 x 6 = 30 Marks)**

		CO	BL
Q-1	How does Design Thinking contribute to solving real-world problems? Illustrate with a suitable approach.	1	2
Q-2	What are Skewness and Kurtosis in data analysis, and why are they important for understanding data distribution?	2	3
Q-3	Explain the concept of Data Integration. What obstacles commonly arise during integration in complex data systems?	1	4
Q-4	What makes a design "good"? List key features of a good design and contrast them with traits of a bad design.	3	3
Q-5	Define Data Science. In what ways is it transforming industries and everyday life?	3	2
Q-6	What is a Heat Map? Where are heat maps most effectively used in data analysis or visualization?	2	4
Q-7	In a Data Science project, what roles are involved, and how do they collaboratively drive the project from start to finish?	2	2
Q-8	Discuss five commonly used data collection methods. How do these support effective data-driven decision-making?	1	3

✓ Q-9	What are the five Vs of Data? Explain their relevance in the context of modern data science applications.	3	4
✓ Q-10	How would you apply the principles of design thinking in the creation of an eco-friendly and sustainable product?	4	3
SECTION-B (LONG ANSWER TYPE QUESTIONS) Instructions: Answer any <i>four</i> questions in detail. Each question carries 10 marks. (4 x 10 = Marks)		CO	BL
✗ Q-1	What is Data Science? Explain its applications in at least five different fields with real-world examples.	1	3
✗ Q-2	Describe the stages of a Data Science project. How do these stages align with the Data Analytics Life Cycle?	2	4
Q-3	Discuss data transformation strategies, define any five strategies in detail.	2	3
✗ Q-4	What is Data Integration? Explain the key challenges faced during integration in large-scale data environments.	1	4
✗ Q-5	Calculate the mean, median, and mode of the following dataset: 24, 22, 34, 112, 12, 145, 56, 78, 90, 34, 21, 67, 90, 55, 23, 50, 34 Also, create a box plot to visually represent the data.	3	4
✗ Q-6	Explain the five stages of the Design Thinking process. Provide examples for each stage to support your explanation.	1	3
Q-7	What are the different ways Design Thinking can be applied for Rural Development? Discuss with practical use cases.	3	5
Q-8	Discuss the following Use cases with respect to application of Data Science and Design thinking for problem solving a) Amazon Go store b) The Good Kitchen case study	2	3
